

MATH 4720/MSSC 5720 Spring 2026 Quiz 4 (Take-home Practice)

1. A one-way ANOVA is conducted to compare mean test scores from 4 tutoring programs. The ANOVA table gives us
 - Between-group sum of squares $SSB = 54$
 - Within-group mean square $MSW = 6$
- (a) Calculate the Between-group mean square MSB , and the F test statistic.
- (b) Suppose an ANOVA test gives a p value of 0.018. Use $\alpha = 0.05$. What decision should be made? State the conclusion in context.